

PathFinder — Project Charter

1. Executive Summary

PathFinder is an enterprise-grade, full-stack web platform engineered to bridge the gap between students seeking academic and professional opportunities and organizations offering them. The platform centralizes the discovery, application, and management of Internships, Scholarships, Grants, and Hackathons into a single, unified ecosystem.

The system incorporates role-based access control, an AI-powered smart matching engine, automated document generation, and a comprehensive administrative dashboard — all built with modern web technologies and deployed on cloud infrastructure for global accessibility.

2. Vision

To become the definitive digital bridge between student talent and organizational opportunity, eliminating fragmentation in the opportunity discovery and application process.

3. Mission

To empower students with intelligent tools that streamline their career development journey, while providing organizations with efficient systems to identify, evaluate, and recruit top talent.

4. Problem Statement

Currently, students face a fragmented landscape when searching for opportunities:

- **Scattered Information:** Opportunities are spread across dozens of websites, university portals, and social media channels with no single source of truth.
 - **Manual Application Tracking:** Students resort to spreadsheets, sticky notes, or memory to track application statuses across multiple platforms.
 - **No Intelligent Matching:** Existing platforms provide basic keyword search but lack intelligent matching based on a student's full academic and skills profile.
 - **Poor Feedback Loops:** Students submit applications and rarely receive meaningful feedback on their fit for a position.
 - **Organization Overload:** Organizations receive hundreds of generic applications with no automated way to assess candidate fit against their specific requirements.
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5. Proposed Solution

PathFinder solves these problems through:

1. **Centralized Opportunity Hub:** A single platform aggregating Internships, Scholarships, Grants, and Hackathons.
 2. **Smart Matching Engine:** AI-powered compatibility scoring between student profiles and opportunity requirements.
 3. **Automated Document Generation:** AI-assisted SOP, Cover Letter, Essay, and Personal Statement generation personalized to each student's profile.
 4. **Real-time Application Tracking:** A timeline-based system showing every status change from submission to decision.
 5. **Administrative Oversight:** A comprehensive admin dashboard with analytics, user management, audit logs, and system health monitoring.
 6. **Support Ticketing System:** Built-in communication channel between users and administrators for issue resolution.
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6. Objectives

ID	Objective	Measurable Outcome
O1	Create a unified opportunity discovery platform	Students can browse, search, and filter opportunities by category, location, and keyword
O2	Implement intelligent student-opportunity matching	Each application receives a 0-100 match score with a human-readable explanation
O3	Build role-based dashboards	Three distinct portal experiences for Students, Organizations, and Admins
O4	Enable AI-powered document generation	Students can generate SOPs, Cover Letters, Essays, and Personal Statements
O5	Deploy to production cloud infrastructure	Frontend on Vercel, Backend on Render with zero-downtime deployments

7. Scope

In Scope

- User Registration and Authentication (JWT-based)
- Student Profile Management (skills, certifications, portfolio links, resume upload)
- Organization Profile Management (company details, logo, industry)
- Opportunity CRUD (Create, Read, Update, Delete) Operations
- Application Submission, Tracking, and Status Management
- AI Match Scoring Engine
- AI Document Generation (SOP, Cover Letter, Essay, Personal Statement)
- Resume PDF Parsing
- Notification System
- Bookmarking System
- Admin Dashboard with Analytics
- Support Ticketing System
- Search and Filter Functionality
- Dark Mode / Light Mode Theme Toggle
- Responsive Design for Desktop and Tablet
- Cloud Deployment (Vercel + Render)

Out of Scope (v1.0)

- Native Mobile Application
- Real-time Chat Between Users
- Payment Processing
- Government API Integration
- Multi-language Support (i18n)
- Video Interviews

8. Stakeholders

Role	Description	Responsibilities
Students	Primary end-users seeking opportunities	Register, build profiles, apply to opportunities, use AI tools
Organizations	Companies and institutions offering opportunities	Post opportunities, review applicants, manage application statuses
Administrators	Platform maintainers	Manage users, monitor analytics, resolve support tickets, ensure platform health
Developers	Engineering team	Build, test, deploy, and maintain the codebase

9. Constraints

- **Budget:** The platform must operate within the free tier of all cloud services (Vercel Hobby, Render Free).
- **Database:** SQLite is used for simplicity and portability, which limits concurrent write operations.
- **AI API:** External AI features depend on API key availability; the system includes mock fallbacks when keys are unavailable.
- **Timeline:** The MVP was developed within a compressed internship timeframe.

10. Assumptions

- Users have modern web browsers (Chrome, Firefox, Edge, Safari) with JavaScript enabled.
- Organizations will self-register and manage their own opportunity postings.
- Students will maintain accurate and up-to-date profiles for the matching engine to function optimally.
- The platform will be accessed primarily from desktop and tablet devices.

11. Risks

Risk	Probability	Impact	Mitigation
AI API rate limits or downtime	Medium	High	Mock fallback responses are built into every AI endpoint
Render free tier cold starts (50s delay)	High	Medium	Documented in user-facing materials; users are informed of initial load times

Database corruption (SQLite file-based)	Low	High	Regular backups; Prisma migrations track schema changes
Security breach via JWT theft	Low	Critical	Tokens expire after 24 hours; passwords are bcrypt-hashed with 10 salt rounds

12. Timeline

Phase	Duration	Deliverables
Phase 1: Foundation	Week 1	Project setup, database schema, authentication, basic CRUD
Phase 2: Core Features	Week 2	Student/Org dashboards, application flow, matching engine
Phase 3: AI Integration	Week 3	Document generation, resume parsing, recommendations
Phase 4: Polish & Deploy	Week 4	Admin dashboard, ticketing system, deployment, documentation

13. Success Criteria

- All three user roles (Student, Organization, Admin) can complete their full workflow end-to-end.
 - The AI matching engine produces scores for every application.
 - The platform is deployed and accessible via public URLs.
 - Another developer can clone the repository and run the project locally using only this documentation.
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14. Deliverables

1. Fully functional web application (Frontend + Backend)
 2. Deployed production environment (Vercel + Render)
 3. Comprehensive documentation suite (this document set)
 4. Database schema and migration files
 5. Developer guide for future contributors
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15. Version History

Version	Date	Author	Changes
v1.0	July 2026	Vaibhav Jain	Initial release — MVP with all core features

End of Project Charter